

Practice Worksheet 1: Computation

CS 101 Algorithmic Problem Solving

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1. Chef's Test

Chef appeared for a placement test.

The test is worth X points and has exactly 10 questions. Each question is worth the same number of points. Chef got N questions correct.

Given X and N , determine the score that Chef will get.

Constraints

- $X, N \in \mathbb{Z}$
- $10 \leq X \leq 200$
- $0 \leq N \leq 10$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and N respectively.

The output must contain a single number denoting the corresponding score earned by Chef.

Sample

Input	Output
20 3	6
15 5	7.5

In the first case, $(X, N) = (20, 3)$. There are 20 points for 10 questions. Each question is therefore worth 2 points. As Chef got 3 questions correct, the score is $3 \times 2 = 6$.

In the second case, $(X, N) = (15, 5)$. There are 15 points for 10 questions. Each question is therefore worth 1.5 points. As Chef got 5 questions correct, the score is $5 \times 1.5 = 7.5$.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
10 3	3
100 10	100
130 4	52

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
12, 7	8.4
14, 5	7
17, 8	13.6

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: X,N

OUTPUT: (X/10)*N

2. Squats

Qasim went to the gym today. He decided to do X sets of squats. Each set consists of N squats.

Given X and N , determine how many squats Qasim did.

Constraints

- $X, N \in \mathbb{Z}$
- $1 \leq X, N \leq 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of X and N respectively.

The output must contain a single number denoting the number of squats Qasim did.

Sample

Input	Output
1 2	2
10 10	100

In the first case, $(X, N) = (1, 2)$. Qasim did 1 set of squats and each set comprised of 2 squats. Therefore, Qasim did 2 squats.

In the second case, $(X, N) = (10, 100)$. Qasim did 10 sets of squats and each set comprised of 10 squats. Qasim did a total of $10 \cdot 10 = 100$ squats.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
10 1	10
24 10	240
9 9	81

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
4, 45	180
3, 6	18
2, 9	18

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: X, N

OUTPUT: $X \cdot N$

3. Where's the lamb biryani?

Michelin starred chef and lamb sauce enthusiast, Gordon Ramsey, wants to learn how to make a perfect biryani. For this reason, he's come to Karachi to learn under the tutelage of Uncle Farhan, owner of the renowned Farhan Biryani restaurant.

Uncle Farhan charges X rupees per class. Chef Ramsey can attend Y classes per week. Uncle Farhan suggests he should attend classes for N weeks in order to become a true master.

Given values of X , Y , and N , determine the total amount of money Chef Ramsey will have to pay.

Constraints

- $X, Y, N \in \mathbb{Z}$
- $1 \leq X, N \leq 10^5$
- $1 \leq Y \leq 7$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of X , Y , and N respectively.

The output must contain a single number denoting the total amount of money Chef Ramsey has to pay.

Sample

Input	Output
500 3 3	4500
1000 7 1	7000

In the first case, $(X, Y, N) = (500, 3, 3)$. Uncle Farhan charges 500 rupees per class and Chef Ramsey has to take a total of 9 classes (3 classes per week for 3 weeks). He has to pay a total of $500 \cdot 9 = 4500$ rupees.

In the second case, $(X, Y, N) = (1000, 7, 1)$. Uncle Farhan charges 1000 rupees per class and Chef Ramsey has to take a total of 7 classes (7 classes per week for 1 week). He has to pay a total of $1000 \cdot 7 = 7000$ rupees.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1350 5 4	27000
500 7 52	182000
1337 3 10	40110

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
23, 3, 4	276
29, 5, 2	290
30, 3, 3	270

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: X,Y,N

OUTPUT: X*Y*N

4. True or False?

Professor Oak has created an exam containing N true or false questions. Each question is worth 1 mark and there is no negative marking.

Ash and Gary are two students taking the exam. In an incredible coincidence, all answers of Ash and Gary are opposite of each other, i.e. if Ash marked a statement "true", Gary marked it "false", and vice versa.

Ash scored K marks on the exam.

Given values of N and K , determine the marks Gary scored.

Constraints

- $N, K \in \mathbb{Z}$
- $1 \leq N \leq 100$
- $0 \leq K \leq N$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of N and K respectively.

The output must contain a single number denoting Gary's obtained marks.

Sample

Input	Output
1 1	0
5 2	3

In the first case, $(N, K) = (1, 1)$. There is only 1 question in the exam and Ash obtained 1 mark, i.e. he answered the question correct. Therefore, Gary would have answered it incorrectly and obtained 0 marks.

In the second case, $(N, K) = (5, 2)$. There are 5 questions in the exam and Ash obtained 2 marks, i.e. he got 2 questions correct and 3 incorrect. Therefore, Gary would have answered 2 questions incorrect and 3 questions correct and obtained 3 marks.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
50 0	50
100 50	50
25 12	13

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
65, 34	31
69, 56	13
23, 7	16

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: N, K

OUTPUT: $N - K$

5. Waiting Time

Chef is eagerly awaiting a piece of information. Her secret agent needs K weeks to obtain the information and share it with Chef.

X days have already passed and Chef is now restless.

Given K and X determine the number of *remaining* days that Chef has to wait before getting the information.

Constraints

- $K, X \in \mathbb{N}$
- $1 \leq K \leq 10$
- $1 \leq X < 7K$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of K and X .

The output must contain a single number denoting the remaining number of days that Chef has to wait.

Sample

Input	Output
1 5	2
2 13	1

In the first case, $(K, X) = (1, 5)$. The information would be revealed after 1 week out of which 5 days have already passed. 2 days remain before the information can be shared.

In the second case, $(K, X) = (2, 13)$. The information would be revealed after 2 weeks out of which 13 days have already passed. 1 days remains before the information can be shared.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
3 16	5
5 7	28
6 12	30

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
3, 7	14
7, 9	40
2, 10	4

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: K, X

OUTPUT: $(K*7)-X$

6. Coconut

You are vacationing in Hawaii which has 2 different types of coconuts, type A and type B . Type A contains x_a millilitres of coconut water and type B contains x_b grams of coconut pulp. You remember that your nutritionist has advised you to consume X_a millilitres of coconut water and X_b grams of coconut pulp every week during your visit.

Given values of x_a , x_b , X_a , and X_b , determine how many total coconuts you will need each week.

Constraints

- $x_a, x_b, X_a, X_b \in \mathbb{Z}$
- $100 \leq x_a \leq 200$
- $400 \leq x_b \leq 500$
- $1000 \leq X_a \leq 1200$
- $1000 \leq X_b \leq 1500$
- x_a divides X_a
- x_b divides X_b

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of x_a , x_b , X_a , and X_b respectively.

The output must contain a single number denoting the total number of coconuts you need in a week.

Sample

Input	Output
100 400 1000 1200	13
150 400 1200 1200	11

In the first case, $(x_a, x_b, X_a, X_b) = (100, 400, 1000, 1200)$. Type A coconuts contain 100 millilitres of coconut water and type B coconuts contain 400 grams of coconut pulp. You need 1000 millilitres of coconut water and 1200 grams of coconut pulp each week. Therefore, you require $\frac{1000}{100} = 10$ type A coconuts and $\frac{1200}{400} = 3$ type B coconuts each week. So a total of 13 coconuts are required per week.

In the second case, $(x_a, x_b, X_a, X_b) = (150, 400, 1200, 1200)$. Type A coconuts contain 150 millilitres of coconut water and type B coconuts contain 400 grams of coconut pulp. You need 1200 millilitres of coconut water and 1200 grams of coconut pulp each week. Therefore, you require $\frac{1200}{150} = 8$ type A coconuts and $\frac{1200}{400} = 3$ type B coconuts each week. So a total of 11 coconuts are required per week.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
100 450 1000 1350	13
200 500 1200 1500	9
150 500 1050 1500	10

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
100, 400, 1000, 1200	13
120, 200, 1200, 2000	20
400, 350, 1200, 700	5

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: x_a, x_b, X_a, X_b

OUTPUT: $(X_a/x_a) + (X_b/x_b)$

7. Chef's Tour

Chef loves to go on tours with her friends.

Chef has arranged N cars that can seat 5 people each and M cars that can seat 7 people each.

Given N and M determine the *maximum* number of people that can travel together in these cars.

Constraints

- $N, M \in \mathbb{Z}$
- $0 \leq N, M \leq 100$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of N and M - the number of 5-seaters and 7-seaters, respectively.

The output must contain a single number denoting the maximum number of people that can travel together in these cars.

Sample

Input	Output
4 8	76
2 13	101

In the first case, $(N, M) = (4, 8)$. There are 4 cars that seat 5 each and 8 cars that seat 7 each. So, $(4 * 5) + (8 * 7) = 76$ people can travel together.

In the second case, $(N, M) = (2, 13)$. There are 2 cars that seat 5 each and 13 cars that seat 7 each. So, $(2 * 5) + (13 * 7) = 101$ people can travel together.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
50 2	264
8 68	516
9 23	206

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
43, 34	453
45, 7	274
22, 11	187

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: N, M

OUTPUT: $(N * 5) + (M * 7)$

8. Weapon division

A warlord has been terribly injured after a battle and it's looking like his time on Earth is coming to an end. Before he departs from the world, he wants to distribute all K of his weapons somewhat fairly among his N sons.

Unfortunately, his math skills have deteriorated greatly after decades of constant warring and bloodshed, and he has asked you for assistance in distributing the weapons. He has set just one condition: the difference between the son that got the least weapons and the son that got the most weapons should be no greater than 1.

Given values of K and N , determine the **minimum** number of weapons a son can have if all K weapons are distributed among them according to the above condition.

Constraints

- $K, N \in \mathbb{Z}$

- $1 \leq K \leq 10^9$
- $1 \leq N \leq 10^5$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the value of K and N respectively.

The output must contain a single number denoting the **minimum** number of weapons a son can possess once all K weapons are distributed according to the warlord's condition.

Sample

Input	Output
8 5	1
5 8	0

In the first case, $(K, N) = (8, 5)$. The warlord has 8 weapons which are to be distributed among 5 sons. The first 5 weapons can be distributed equally among the 5 sons. The remaining 3 weapons will then be given to any 3 sons. Therefore, 3 sons will receive 2 weapons and 2 sons will receive 1 weapon. Hence, minimum number of weapons possessed by a son is 1.

In the second case, $(K, N) = (5, 8)$. The warlord has 5 weapons which are to be distributed among 8 sons. The 5 weapons can be given to any 5 sons equally. The remaining 3 sons will receive no weapons. Hence, the minimum number of weapons possessed by a son is 0.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
15 5	3
39 10	3
24 25	0

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
45, 5	9
34, 36	0
23, 8	2

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: K,N

OUTPUT: K/N if the answer in decimals round down.

9. Group Assignment

Your professor is planning on giving the class a group assignment. There are $2N$ students in your class, with roll numbers from 1 to $2N$. Your roll number is X .

The professor has decided to create N groups of 2 students each. The groups are to be created as follows: the first group contains roll numbers 1 and $2N$, the second group contains roll

numbers 2 and $2N - 1$, the third group contains roll numbers 3 and $2N - 2$, and so on, with the N th group containing roll numbers N and $N + 1$.

You want to find out who your partner will be beforehand.

Given values of N and X , determine the roll number of your partner.

Constraints

- $N, X \in \mathbb{Z}$
- $1 \leq N \leq 10^8$
- $1 \leq X \leq 2N$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the value of N and X respectively.

The output must contain a single number denoting the roll number of your partner.

Sample

Input	Output
2 2	3
3 1	6

In the first case, $(N, X) = (2, 2)$. There are $2 \cdot 2 = 4$ students in the class and your roll number is 2. The groups will be as follows: $\{(1, 4), (2, 3)\}$. You are partnered with roll number 3.

In the second case, $(N, X) = (3, 1)$. There are $2 \cdot 3 = 6$ students in the class and your roll number is 1. The groups will be as follows: $\{(1, 6), (2, 5), (3, 4)\}$. You are partnered with roll number 6.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
15 5	26
35 23	48
123 189	58

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
14, 7	22
44, 4	85
372, 3	742

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: N, X

OUTPUT: $2N+1-X$

10. Handshake

At the annual meeting of Board of Directors of Acme Inc, each attendee shakes hands exactly once with every other attendee.

The Chief Health Officer is interested to know, given a attendees, how many handshakes take place.

Constraints

- $a \in \mathbb{N}$
- $1 \leq a \leq 1000$

Interaction

The input comprises a single line containing an integer denoting the value of a .

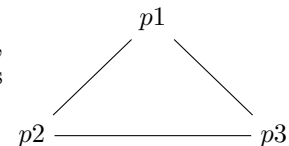
The output must contain a single number denoting the total number of handshakes.

Sample

Input	Output
1	0
3	3

In the first case, there are $a = 1$ attendees. There is no one for the lone attendee to shake hands with. The total number of handshakes is 0.

In the second case, there are $a = 3$ attendees. Let us call them $p1, p2$, and $p3$. The diagram on the right represents the handshakes as edges between two people. The total number of handshakes is 3.



Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2	1
9	36
12	66

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
45	990
33	528
10	45

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

INPUT: a

OUTPUT: $[(a-1)a]/2$

Rough Work